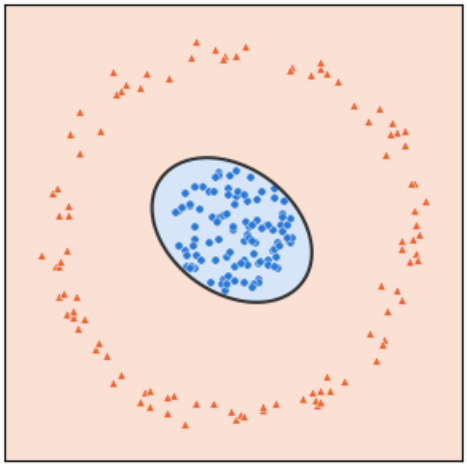
Introduction to Machine Learning

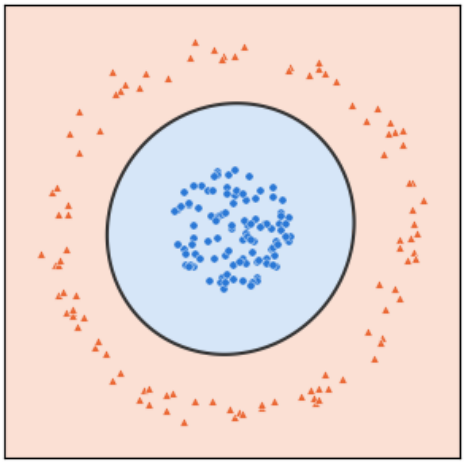
Session 7 Lecture Notes

Kernel Methods II - SVMs

Why do we need a best boundary in contrast to just boundary?

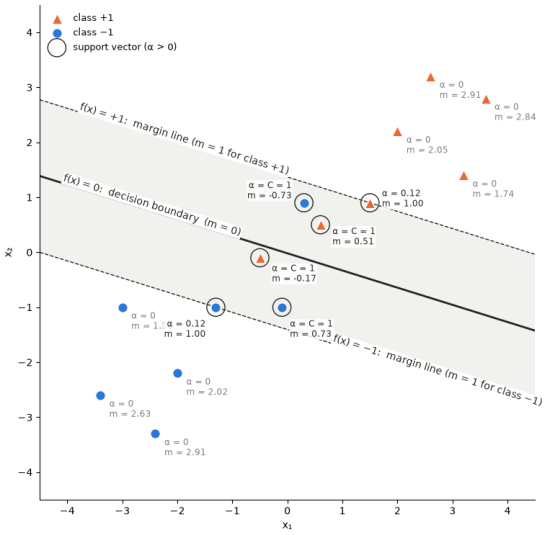


First encountered boundary



Maximum margin boundary

- Maximum-margin boundary means that small noise points are safe in the classification process
- Maximum-margin boundary is a well-defined optimization problem with a unique answer
- The solution depends not on all points in the dataset, but a selected 'important' points — hence the name Support Vector Machine.



$w \cdot x + b = 0$ *is the boundary*

let's fix $\|w\| = 1$

the margin we want to maximalize:

$$y_i(w \cdot x + b) \geq \gamma, \quad y_i \in \{-1, 1\}$$

$$y_i\left(\frac{w}{\gamma}\cdot x + \frac{b}{\gamma}\right) \geq 1$$

$$\|w'\| = \frac{1}{\gamma}\cdot\|w\|, \qquad \|w\| = 1)$$

$$\|w'\| = 1/\gamma$$

$$\frac{1}{\|w'\|} = \gamma$$

The Primal Problem

Maximizing the margin γ is equal to maximizing $\frac{1}{\|w'\|}$, and in practice we minimize $\frac{1}{2}\|w'\|^2$.

Let's just rename w' to w after this point.

Dual problem

- Payoff 1: Kernel compatible
- Payoff 2: Sparsity – the dual problem gives α_i weights per point, which will be minimized so that only some will be important in the calculations

Lagrangian Dual Problem

$$L(w,b,\alpha) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{n}\alpha_i[\, y_i(w\cdot x_i + b) - 1\,]$$

with constraints:

$$\alpha_i \geq 0$$

with derivatives set to 0:

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{n}\alpha_i y_i x_i$$

$$\frac{\partial L}{\partial b} = -\sum_{i=1}^{n}\alpha_i y_i$$

$$w = \sum_{i=1}^{n}\alpha_i y_i x_i \quad and \quad \sum_{i=1}^{n}\alpha_i y_i = 0$$

Then:

$$L = \frac{1}{2}w\cdot w - \sum_{i=1}^{n}\alpha_i[\, y_i(w\cdot x_i + b) - 1\,] =$$

$$= \frac{1}{2}w\cdot w - \sum_{i=1}^{n}[\alpha_i y_i(w\cdot x_i) + \alpha_i y_i b - \alpha_i] =$$

$$= \frac{1}{2}w\cdot w - \sum_{i=1}^{n}\alpha_i y_i(w\cdot x_i) - \underbrace{\sum_{i=1}^{n}\alpha_i y_i}_{=0} b + \sum_{i=1}^{n}\alpha_i =$$

$$= \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j(x_i\cdot x_j) - \sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j(x_i\cdot x_j) + \sum_{i=1}^{n}\alpha_i = -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j\underbrace{(x_i\cdot x_j)}_{K(x_i,x_j)} + \sum_{i=1}^{n}\alpha_i$$

Since this is dual problem, then with w growing α gets smaller. We want to minimize w so we maximize α.

Maximize, subject to $\alpha_i \geq 0$ and $\Sigma\alpha_i y_i = 0$

Now to calculate b, we take points exactly on the margins:

$$y_k f(x_k) = 1$$

$$1 = y_k\left(\sum_{i=1}^{m}[\alpha_i y_i K(x_i, x_k)] + b\right)$$

$$y_k = \sum_{i=1}^{m}[\alpha_i y_i K(x_i, x_k)] + b$$

$$b = y_k - \sum_{i=1}^{m}[\alpha_i y_i K(x_i, x_k)]$$

And so, prediction becomes:

$$f(x) = \sum_{i=1}^{n}[\alpha_i y_i K(x_i, x_k)] + b \qquad \begin{cases} f(x) > 0, & y = 1 \\ f(x) < 0, & y = -1 \end{cases}$$

Then solving, training the SVMs:

SMO (Sequential Minimal Optimization)

Starts with $\alpha = 0$ and picks two α's that violate KKT conditions the most, solve the tiny problem, and iterate until can't improve no more.

Soft Margin

Noisy, overlapping data.

$$y_i(w \cdot x_i + b) \geq 1 - \xi_i$$

- $\xi_i = 0$ if correct position
- $0 < \xi_i \leq 1$ if in the margin
- $\xi_i > 1$ misclassified

We set C as a price of violation.

- Large C — small margin
- Small C — big margin

So we minimize:

$$L = \frac{1}{2}\|w\|^2 + C \cdot \sum_{i=1}^{n}\xi_i$$

with Lagrangian:

$$L = \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i[y_i(w \cdot x_i + b) - 1 + \xi_i] - \mu_i\xi_i$$

$$\frac{\partial L}{\partial \xi} = C - \alpha_i - \mu_i \overset{=0}{\rightarrow} \quad \alpha_i = C - \mu_i, \quad \mu_i \geq 0 \rightarrow \alpha_i \leq C$$

So, then we maximize:

$$\sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

subject to $0 \leq \alpha_i \leq C$ *and* $\Sigma \alpha_i y_i = 0$

Margin of a Point

$$m = y \cdot f(x)$$

so that our Loss function → hinge loss → can be written with a constraint

$$\xi_i = max(0, 1 - m_i)$$

total loss:

$$L = \sum_{i=1}^{n} max(0, 1 - y_i f(x_i))$$

The soft margin m_i is classified into 3 classes, which form the KKT table:

m	α	Type
> 1	0	Not support vector (correct classification)
$= 1$	$0 < \alpha < C$	A margin support vector (on margin, correctly classified)
< 1	$\alpha = C$	Support vector (inside margin or incorrectly classified)

Hyper parameters for SVMs

From algorithm's perspective:

- C

From kernel's perspective:

- γ or C_o, and degree (d)

Multiclass classifiers

- OvR — one vs. rest
- OvO — one vs. one

Binary classifiers for different combinations of classes.